

$$\begin{array}{c}
\frac{}{B\mathcal{R}B \vdash B\mathcal{R}B, B \Vdash A} \text{identity} \quad \frac{}{B \Vdash A, B\mathcal{R}B \vdash B \Vdash A} \text{identity} \\
\hline
\frac{}{B\mathcal{R}B \vdash B\mathcal{R}B, B \Vdash A} \text{identity} \quad \frac{}{B \Vdash A, B\mathcal{R}B \vdash B \Vdash A} \text{identity} \\
\hline
\frac{(B\mathcal{R}B) \rightarrow (B \Vdash A), B\mathcal{R}B \vdash B \Vdash A}{\forall C.((B\mathcal{R}C) \rightarrow (C \Vdash A)), B\mathcal{R}B \vdash B \Vdash A} \text{forallL} \\
\hline
\frac{}{B \Vdash \Box A, B\mathcal{R}B \vdash B \Vdash A} \text{mBoxL} \\
\hline
\frac{}{B\mathcal{R}B \vdash (B \Vdash \Box A) \rightarrow (B \Vdash A)} \text{implyR} \\
\hline
\frac{}{B\mathcal{R}B \vdash B \Vdash (\Box A) \rightarrow A} \text{mImPLYR} \\
\hline
\frac{}{\forall w.(w\mathcal{R}w) \vdash B \Vdash (\Box A) \rightarrow A} \text{forallL} \\
\hline
\frac{}{\forall w.(w\mathcal{R}w) \vdash \forall u.(u \Vdash (\Box A) \rightarrow A)} \text{forallR} \\
\hline
\frac{}{\vdash (\forall w.(w\mathcal{R}w)) \rightarrow (\forall u.(u \Vdash (\Box A) \rightarrow A))} \text{implyR}
\end{array}$$